Status	Finished
Started	Friday, 9 October 2026, 11:56 AM
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Duration	2 mins 45 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It complicates the analysis process
- c. It has no effect on the analysis process
- d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Merge Sort
- b. Quick Sort
- c. Bubble Sort
- d. Insertion Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Bubble Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution
b. Selection
c. Extraction
d. Insertion

The correct answer is:
Insertion

Question **5**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort
b. It is based on the divide-and-conquer approach
c. It does not require recursion
d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It compares adjacent elements and swaps them if necessary
b. It repeatedly finds the minimum element and moves it to the sorted part of the list
c. It divides the list into two halves, sorts each half, and then merges them
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **7**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 40,80,90,150,200,300,400,470,70,120
- c. 200,470,80,150,40,90,300,400,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **8**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. arrange()
- b. sort()
- c. sorted()
- d. order()

The correct answer is: sorted()

Question **9**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Dynamic programming
- b. Greedy method
- c. Divide and conquer
- d. Backtracking

The correct answer is:
Divide and conquer

Question **10**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others
b. Combining sorted sublists
c. Dividing the input into smaller subproblems
d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Heap Sort
b. Bubble Sort
c. Merge Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Selection sort
b. Merge sort
c. Quick sort
d. Heap sort

The correct answer is:
Merge sort

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items
b. Sorting is rarely used in programming
c. Sorting helps in finding duplicates quickly
d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **14**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process
b. It complicates the selection of items
c. It makes it easier to select items based on their relationship to the rest of the items
d. It makes the data unsorted

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **15**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It sorts data out of the box efficiently
b. It requires external libraries
c. It only works with integer arrays
d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently